

PrismIQ Analytics Report

Dataset Intelligence & Data Quality Summary

Generated By:	PrismIQ
Generated At:	2026-07-16 16:59 UTC
Dataset Name:	customeriq demo dataset
Dataset Version:	Version 2
Quality Score:	100.00/100
Report ID:	bc3c4a6d-4cef-485c-8451-efcf3075e450
User ID:	bbfca9e2-788f-4419-8ace-5999bd3cb798
Engine Version:	1.0
Report Schema:	1.0

Executive Summary

The dataset 'customeriq demo dataset' currently holds a quality health score of **100.00/100**. The active version contains a total of 0 missing data cells and 0 duplicate rows. This summary highlights structural anomalies, cleaning lineages, and AI-recommended preprocessing steps.

Dataset Information

Dataset Name:	customeriq demo dataset
Original Filename:	customeriq_demo_dataset.csv
Description:	No description provided
File Size (Bytes):	2616
Import Timestamp:	2026-07-16 16:20:58 UTC

Quality Metrics & Schema Profile

Column Name	Data Type	Missing	Null %
CustomerID	numeric	0	0.00%
InvoiceNo	text	0	0.00%
PurchaseDate	datetime	0	0.00%
CustomerName	categorical	0	0.00%
Email	categorical	0	0.00%
City	categorical	0	0.00%
Country	categorical	0	0.00%
Product	text	0	0.00%
Category	categorical	0	0.00%
Quantity	numeric	0	0.00%
UnitPrice	numeric	0	0.00%
TotalAmount	numeric	0	0.00%

Cleaning Summary & History

Removed Rows	Duplicates	Missing Handled	Outliers Handled	Time
0	0	0	0	7 ms

Exploratory Data Analysis (EDA)

Column	Mean	Median	Std Dev
CustomerID	10008.75	10008.50	5.56
Quantity	3.20	2.00	4.53
UnitPrice	92.95	54.50	103.67
TotalAmount	130.82	102.47	96.13

AI Insights Observations

[OUT_003] Category: Risk		Severity: HIGH
Observation: Extraneous value outliers detected in numeric variables.		Confidence: 0.92%
Root Cause: Outliers were discovered beyond standard 1.5x Interquartile Ranges (IQR), specifically in: `Quantity` (3 outliers, 15.0%), `UnitPrice` (3 outliers, 15.0%), `TotalAmount` (1 outliers, 5.0%).		
[CORR_004] Category: Correlation		Severity: HIGH
Observation: Strong linear relationship detected between numeric columns.		Confidence: 0.95%
Root Cause: Pearson correlation analysis indicates strong associations: `UnitPrice` & `TotalAmount` (r=0.91). Highly correlated variables move together systematically.		
[CARD_005] Category: Statistics		Severity: LOW
Observation: High cardinality in categorical data columns.		Confidence: 0.9%
Root Cause: Categorical parameters contain high ratios of unique values: `InvoiceNo` (20 unique, 100.0%), `CustomerName` (18 unique, 90.0%), `Email` (18 unique, 90.0%), implying they represent textual logs, indexes, or identifiers.		
[TEMP_006] Category: Timeline		Severity: INFO
Observation: Temporal timelines coverage loaded successfully.		Confidence: 0.95%
Root Cause: Datetime intervals spans detected: `PurchaseDate` spanning 54.0 days (from 2025-01-05T00:00:00 to 2025-02-28T00:00:00).		

Actionable Recommendations

Observation Link	Recommended Action	Actionability
Risk	Evaluate outliers. Outliers can represent measurement faults or rare valid events. Consider applying IQR trimming to isolate distribution metrics.	85%
Correlation	These variables exhibit high linear dependency. Consider removing one column or performing dimensionality reduction to avoid multicollinearity in downstream models.	70%
Statistics	Ensure these columns are standard strings. Consider group standardizations or mapping key features.	60%
Timeline	If running cyclic or seasonal forecasts, ensure this timeline span covers at least one complete calendar year or cyclic sales window.	50%

Appendix & Methodology Documentation

Quality Score Index is calculated as a weighted average of missing cell percentages and duplicate row ratios. Missing weights and duplicate weights default to 50.0% each, adjusted via server configuration parameters.

Engine Version: 1.0.0 | **Report Schema:** 1.0 | **Checksum Algorithm:** SHA-256